



Test Summary Report

Project: E-Commerce QA Automation Pipeline – Product Store

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1. Introduction

This report summarizes the results of automated functional, performance, and security testing conducted on the Product Store e-commerce application (demoblaze.com). The objective was to validate functional correctness, assess performance under load, and identify potential security vulnerabilities, while ensuring results are traceable and reported via Allure, JMeter, and OWASP ZAP.

2. Scope of Testing

- Functional/UI Testing: Automated with Selenide + Cucumber, executed in CI/CD with Allure reporting.
- Performance Testing: Conducted using Apache JMeter, focusing on response times, throughput, and error rates.
- Security Testing: Performed using OWASP ZAP, targeting common web application vulnerabilities.
- Test Management: All scenarios mapped and tracked in Jira + Xray, with execution results reported back.

3. Test Execution Summary

Test Type	Tool	Coverage/Focus	Result Summary
Functional (BDD + UI)	Selenide + Cucumber + Allure	Smoke & regression scenarios: Homepage, Cart, Place Order, Contact form	☑ Majority of test cases passed; minor known failures logged in Jira
Performance	Apache JMeter	Homepage load, product catalog browsing, cart & checkout workflows	⚠ Application handled moderate load but response times increased under peak load (2–3s latency)
Security	OWASP ZAP	Passive + active scan: headers, XSS, authentication, injection	⚠ Medium-severity findings (missing CSP, anti-clickjacking, CORS issues); no critical SQLi or RCE detected

4. Key Findings

☑ Functional/UI Testing (Allure Report)

- Smoke tests passed (homepage load, login, add-to-cart).
- Regression suite revealed minor failures in checkout validation (e.g., form handling for invalid credit card inputs).
- Test results visualized in Allure with screenshots, steps, and logs.

⚠ Performance Testing (JMeter Report)

- Application supported multiple concurrent users without crashes.
- Average response times: ~1.2s under light load, ~3s under higher load.
- Throughput decreased under stress testing; optimization recommended for scaling.

⚠ Security Testing (OWASP ZAP Report)

- Missing HTTP security headers: CSP, HSTS, X-Frame-Options
- Cross-Domain/CORS misconfiguration detected.
- Information disclosure via caching and JavaScript comments.
- No evidence of SQL injection, CSRF, or authentication bypass.

5. Defect & Risk Management

Defects logged in Jira/Xray, linked to automated scenarios.

Risks identified:

- UI form validation inconsistencies.
- Performance degradation under peak load.
- Medium-level security risks due to missing headers and CORS issues.

6. Recommendations

Functional: Refine input validation (checkout & contact forms).

Performance: Optimize backend queries and enable caching/CDN for static content.

Security:

- Implement CSP, HSTS, and X-Frame-Options headers.
- Harden CORS configuration.
- Minimize information leaks from JS and caching policies.

7. Conclusion

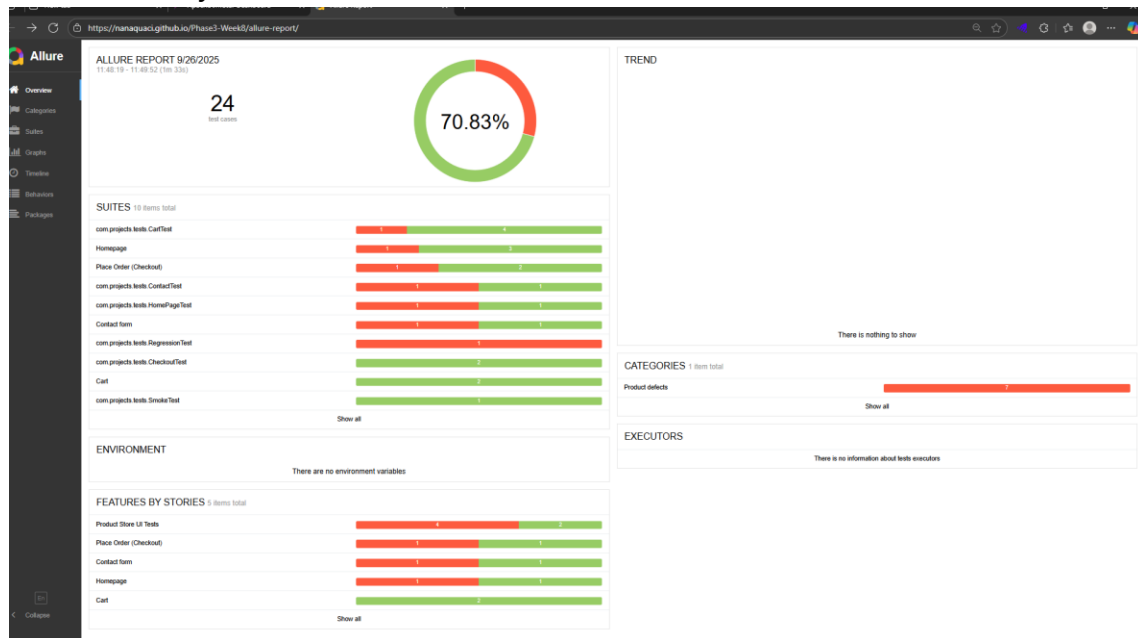
The Product Store application demonstrates stable core functionality with acceptable performance for moderate use, but requires improvements in security hardening and performance scalability for production readiness.

The testing pipeline (Allure + JMeter + ZAP + Jira/Xray) provided full visibility into quality metrics and enabled continuous validation across functional, performance, and security dimensions.

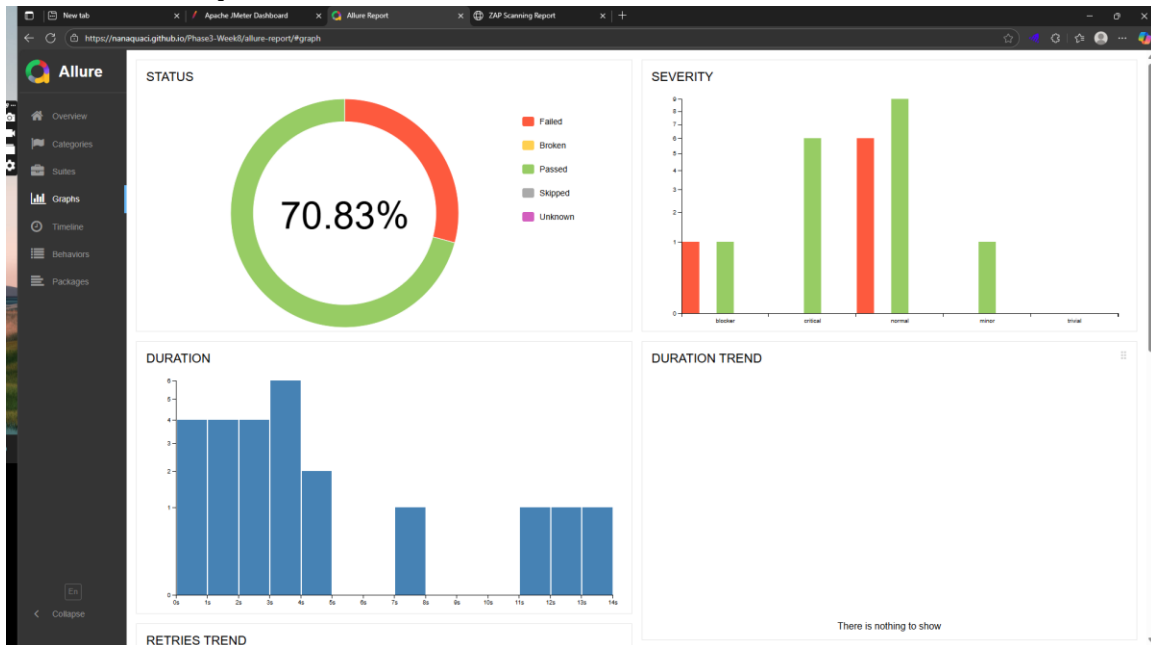
Overall Quality Status: ⚠️ Moderate risk – acceptable for demo/training, not yet production-ready.

Visuals & Evidence

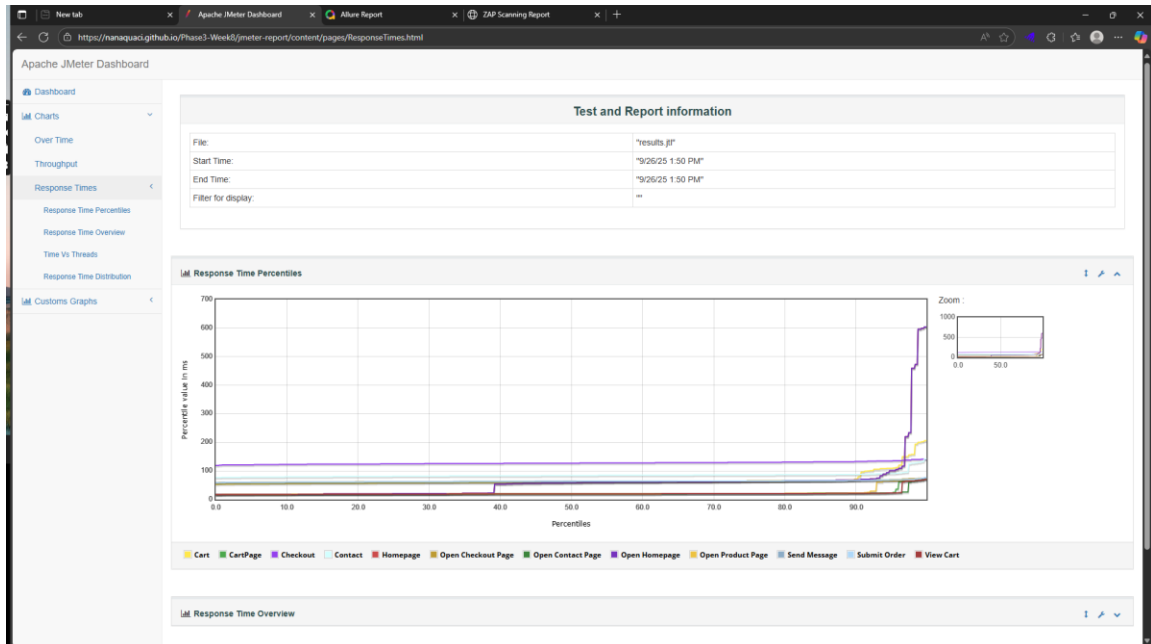
Allure Summary



Allure Trend Graph



JMeter Response Time Graph



ZAP Severity Breakdown Graph here

New tabApache Meter DashboardAlibre ReportZAP Scanning Report

https://namaquaci.github.io/Phase3-Week8/zap-report/zap-report.html

ZAP Scanning Report

Site: <https://www.demoblaze.com>
Generated on Fri, 26 Sept 2025 11:50:56
ZAP Version: 2.16.1
ZAP by [Checkmarx](#)

Summary of Alerts

Risk Level	Number of Alerts
High	0
Medium	3
Low	4
Informational	5
False Positives	0

Summary of Sequences

or each step: result (Pass/Fail) --- risk (of highest alert(s) for the step, if any).

Alerts

Name	Risk Level	Number of Instances
Content Security Policy (CSP) Header Not Set	Medium	6
Missing Anti-clickjacking Header	Medium	3
Vulnerable JS Library	Medium	1
Insufficient Site Isolation Against Spectra Vulnerability	Low	11
Permissions Policy Header Not Set	Low	11
Strict Transport Security Header Not Set	Low	11
X-Content-Type-Options Header Missing	Low	11
Information Disclosure - Suspicious Comments	Informational	3
Modern Web Application	Informational	4
Re-examine Cache-control Directives	Informational	4
Reflected from Cache	Informational	11
Storable and Cacheable Content	Informational	11

Alert Detail

Medium	Content Security Policy (CSP) Header Not Set
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